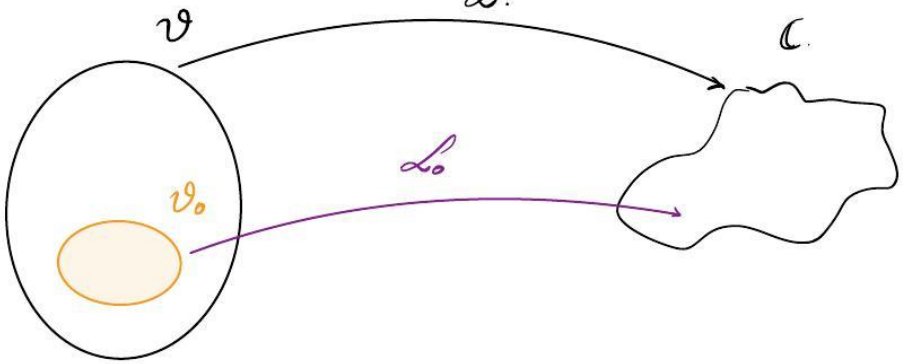
Clase 6 Análisis Numérico

Teorema de aproximación de Weierstrass: Toda función continua en un compacto puede ser aproximada uniformemente por polinomios. i.e, si $f \in C(I)$ y $\varepsilon > 0$ entonces $\exists p \in P$ t.q $\|f-p\|_\infty < \varepsilon$.

Teorema de Hahn-Banach: Sea $\mathcal{V}$ un espacio normado complejo y $\mathcal{L}_0 : \mathcal{V}_0 \longrightarrow \mathbb{C}$ es continua sobre el subespacio $\mathcal{V}_0 \subseteq \mathcal{V}$ entonces $\exists \mathcal{L} : \mathcal{V} \longrightarrow \mathbb{C}$ t.q $\mathcal{L}|_{\mathcal{V}_0} = \mathcal{L}_0$ y $\|\mathcal{L}\| = \|\mathcal{L}_0\|$.



Recordemos que

$$\|\mathcal{L}\| = \inf\{c \mid |\mathcal{L}x| \le c\|x\|\}$$
$$= \left\{c \mid \left|\mathcal{L}\left(\frac{x}{\|x\|}\right)\right| \le c\right\}$$
$$= \sup_{\|x\|=1} \{|\mathcal{L}x|\}.$$

(Spanning Criterion): Sea W un subespacio de $\mathcal{V}$ donde $\mathcal{V}$ es normado entonces $v \in \overline{W}$ si y solo si: $\forall \mathcal{L} \in W^\perp \ \mathcal{L}(v) = 0$.

$$W^\perp = \{\mathcal{L} : W \to \mathbb{C} \text{ continuous} \mid \mathcal{L}(w) = 0 \ \forall w \in W\}$$

Proof. [$\Rightarrow$] Sea $v \in \overline{W}$ y $\mathcal{L} \in W^\perp$ es decir $\mathcal{L}|_W \equiv 0$ Sea $\varepsilon > 0$ entonces existe $w \in W$ t.q $\|v-w\| < \varepsilon$.

En tonces

$$\|\mathcal{L}(v)\| = \|\mathcal{L}(v-w)\|$$
$$\le \|\mathcal{L}\|\|v-w\|$$
$$< \|\mathcal{L}\|\varepsilon.$$

$$\|\mathcal{L}\| = \inf\{c \mid \|\mathcal{L}x\| \le c\|x\|\}$$
$$= \sup\left\{\frac{\|\mathcal{L}x\|}{\|x\|} ; x \neq 0\right\}$$
$$\Rightarrow |\mathcal{L}x\| \le \|\mathcal{L}\|\|x\|$$

dado que $\mathcal{L}$ es continua i.e, $\|\mathcal{L}\| < \infty$ y $\varepsilon > 0$ arbitrario, $\mathcal{L}(v) = 0$.

[⇐]. Por contrarecíproco:

$v \notin \overline{W}$ entonces $\exists \mathcal{L} \in W^{\perp}$ $\mathcal{L}(v) \neq 0$

Supongamos que $v \notin \overline{W}$ entonces $d(v, W) > 0$. Definamos:

$V_0 = \{av + w \mid w \in W, a \in \mathbb{C}\}$. es subespacio de V.

y definamos $\mathcal{L}_0 : V_0 \longrightarrow \mathbb{C}$. Note que $\mathcal{L}_0|_W \equiv 0$.

$av + w \longmapsto a$

Claim: $\mathcal{L}_0$ es continuo

↪ $\|\mathcal{L}\| = \inf\{c \mid |\mathcal{L}x| \leq \|x\| c\}$
↪ $\mathcal{L}$ es acotado si $\exists K$ $\|\mathcal{L}x\| \leq K\|x\|$

$\|av + w\| \geq |a| d(v, W)$ entonces $|a| \leq \dfrac{\|av + w\|}{d(v, W)}$ entonces.

$|\mathcal{L}_0(av + w)| = |a| \leq \|av + w\| \cdot C$ donde $C = \dfrac{1}{d(v, W)}$

por definición $\mathcal{L}_0$ es acotado y por tanto continuo.

Por thm Hahn-Banach $\exists \mathcal{L} : V \to \mathbb{C}$ t.q $\mathcal{L}|_{V_0} = \mathcal{L}_0$ y $\|\mathcal{L}\| = \|\mathcal{L}_0\|$.

$\mathcal{L}|_W = \mathcal{L}_0|_W \equiv 0$ es decir $\mathcal{L} \in W^{\perp}$ y $\mathcal{L}(v) = \mathcal{L}_0(v) = 1 \neq 0$. ∎

Proof: (Thm de Weierstrass): (WAT). Es suficiente probar el teorema para $f(x) = |x|$ for $-1 \leq x \leq 1$. Sea $V = (C[-1, 1], \|\cdot\|)$. y.

$W = \{\text{polinomios}\}$

Goal: $|x| \in \overline{W}$ Por Spanning Criterion $\forall \mathcal{L} \in W^{\perp}$ $\mathcal{L}(f) = 0$.

$f(x) = |x|$

Sea $\mathcal{L} \in W^{\perp}$ Vamos a probar que $\mathcal{L}(|x|) = 0$.

Sea $|t|^z$, $z = x + iy$ y $x > 0$ entonces $|t|^z = e^{z \ln |t|}$

y consideremos $f(z) := \mathcal{L}(|t|^z)$ queremos que $f(1) = 0$.

Lemma: $f(x, y) = \mathcal{L}(|t|^{x+iy})$ es holomorfa en $U = \{z \mid \mathrm{Re}\, z > 0\}$.

(f es holomorfa si: $\dfrac{\partial f}{\partial \bar{z}} = 0$).

Proof Lemma: Definimos $g(x,y,t) = |t|^{x+iy} = e^{(x+iy)\ln|t|}$

$g : U \times \underbrace{[-1,1]}_{K} \longrightarrow \mathbb{C}$ entonces $g_{x,y} : [-1,1] \longrightarrow \mathbb{C}$.

$f(x,y) = \mathcal{L}(g_{x,y})$.

Derivando:

$$\left.\begin{aligned} \frac{\partial}{\partial x}(g) &= |t|^{x+iy}\ln|t| \\ \frac{\partial}{\partial y}(g) &= i\,|t|^{x+iy}\ln|t| \end{aligned}\right\} \text{ son continuas.}$$

entonces:

$$\frac{\partial f}{\partial x} = \mathcal{L}\left(\frac{\partial}{\partial x}|t|^{x+iy}\right) = \mathcal{L}\left(|t|^{x+iy}\ln|t|\right)$$

$$\frac{\partial f}{\partial y} = \mathcal{L}\left(\frac{\partial}{\partial y}|t|^{x+iy}\right) = i\,\mathcal{L}\left(|t|^{x+iy}\ln|t|\right)$$

Así

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right) = 0.$$ Así f es holomorfa en U ∎

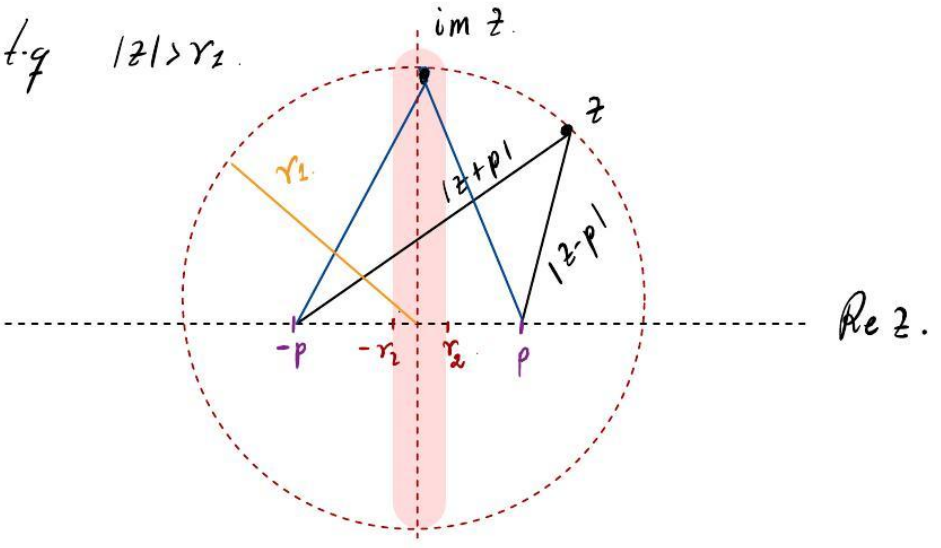
Dado que $f(2n) = 0$ entonces:

$$\frac{f(z)}{z-2}, \frac{f(z)}{z-4}, \cdots, \frac{f(z)}{z-2n} \text{ son todas holomorfas.}$$

Lemma: Sea $p \in \mathbb{R}$, $p > 0$ y $\varepsilon > 0$ Entonces $\exists r_1, r_2 > 0$ t.q.

$$\left|\frac{z+p}{z-p}\right| \leq 1+\varepsilon. \quad \forall z \in \mathbb{C} \text{ t.q } |z| > r_1$$

o $|\mathrm{Re}\, z| < r_2$



Consideremos

$$\frac{z+1}{z-1}\cdot\frac{z+2}{z-2}\cdot\cdots\cdot\frac{z+N}{z-N}\,f(z) \quad \text{es holomorfa en } \mathcal{U}.$$

Entonces $\forall \varepsilon>0$ $\exists r_1, r_2>0$ s.t $1\le K\le N$ $\left|\frac{z+K}{z-K}\right| \le 1+\varepsilon$. para $|z|>r_1$ o $|Re\, z|<r_2$.

$$\left|\frac{z+1}{z-1}\cdot\cdots\cdot\frac{z+N}{z-N}\right| |f(z)| \le (1+\varepsilon)^N |f(z)|$$

Así

$$|f(z)| = |\mathcal{L}(|t|^z)| \le \|\mathcal{L}\|\,\||t|^z\|$$

$$= \|\mathcal{L}\| \max\{|t|^z ;\ -1\le t\le 1\} = \|\mathcal{L}\|.$$

así $\left|\frac{z+1}{z-1}\cdot\cdots\cdot\frac{z+N}{z-N}\right| |f(z)| \le \underbrace{(1+\varepsilon)^N}_{\varepsilon\to 0}\|\mathcal{L}\| \le \|\mathcal{L}\|$

Note que

$$\left|\frac{z+K}{z-K}\right| = \left|1+\frac{2z}{K-z}\right| \quad \text{entonces}$$

$$\left|1+\frac{2z}{2-z}\right|\cdots\left|1+\frac{2z}{2N-z}\right| |f(z)| \le \|\mathcal{L}\|\,(1+\varepsilon)^N \le \|\mathcal{L}\|$$

Si $z=1$.

$$|f(1)|\left(1+\tfrac{2}{1}\right)\left(1+\tfrac{2}{3}\right)\cdots\left(1+\frac{2}{2N-1}\right) \le \|\mathcal{L}\|(1+\varepsilon)^N \le \|\mathcal{L}\|$$

Entonces. $\left(1+\sum_{i=1}^{N}{}'\frac{2}{2i-1}\right)|f(1)| \le \left(1+\frac{2}{1}\right)\left(1+\frac{2}{3}\right)\cdots\left(1+\frac{2}{2N-1}\right)|f(1)|$.

$$\le \|\mathcal{L}\|$$

tomando $N\to\infty$. $\sum_{i=1}^{\infty}{}'\frac{2}{2i-1}=\infty$. entonces. $f(1)=0$.

lo que prueba que $|\mathcal{L}(|t|)|=0$. ∎

Maximum - Modulus thm: Sea $\mathcal{U}\subseteq\mathbb{C}$ conjunto abierto y acotado y f holomorfa en un conjunto abierto que contiene a $\mathcal{U}$. Si $|f(z)|\le C$ $\forall z\in\partial\mathcal{U}$ entonces $|f(z)|\le C$ $\forall z\in\mathcal{U}$.

i.e, $\forall z \in U \quad |f(z)| \leq \max\{|f(z)| \mid z \in \partial U\}$

Problema: $\forall \{a_i\}$ sucesión de no negativos y $\forall n \in \mathbb{N}$.

$$1 + \sum_{i=1}^{n} a_i \leq \prod_{i=1}^{n} (1+a_i).$$

Thm: Sea $\{\Phi_0, \Phi_1, \dots, \Phi_n, \dots\}$ es un conjunto de polinomios ortogonales Entonces $\exists\ (\alpha_k)_{k=1}^{\infty}, (\beta_k)_{k=1}^{\infty}, (\gamma_k)_{k=1}^{\infty}$ tal que se cumple la fórmula de recurrencia

$$\Phi_{k+1}(x) = \alpha_k (x - \beta_k)\Phi_k(x) - \gamma_k \Phi_{k-1}(x)$$

Proof: Note que $x\Phi_k \in \Pi_{k+1}$ entonces $\{\Phi_0, \Phi_1, \dots, \Phi_{k+1}\}$ es base ortogonal. así $\left\{\frac{\Phi_0}{\|\Phi_0\|}, \frac{\Phi_1}{\|\Phi_1\|}, \dots, \frac{\Phi_{k+1}}{\|\Phi_{k+1}\|}\right\}$ es base ortonormal.

$$x\Phi_k = \sum_{i=0}^{k+1} \left\langle x\Phi_k, \frac{\Phi_i}{\|\Phi_i\|} \right\rangle \cdot \frac{\Phi_i(x)}{\|\Phi_i\|}$$

$$= \sum_{i=0}^{k+1} \frac{\langle x\Phi_k, \Phi_i \rangle}{\|\Phi_i\|^2} \Phi_i(x)$$

Note que $x\Phi_i \in \Pi_{i+1} \Rightarrow \sigma_{k,i} = 0$ para todo $i = 0, 1, \dots, k-2$.

$$x\Phi_k(x) = \frac{\langle x\Phi_k, \Phi_{k+1} \rangle}{\|\Phi_{k+1}\|^2} \Phi_{k+1}(x) + \frac{\langle x\Phi_k, \Phi_k \rangle}{\|\Phi_k\|^2} \Phi_k(x) + \frac{\langle x\Phi_k, \Phi_{k-1} \rangle}{\|\Phi_{k-1}\|^2} \Phi_{k-1}(x)$$

Lemma: Suponga que $\{\Phi_0, \Phi_1, \dots, \Phi_k, \dots\}$ son polinomios ortogonales para $\langle \cdot, \cdot \rangle$. Entonces $\langle \Phi_k, q \rangle = 0 \quad q \in \Pi_{k-1}$. $\{\text{degree}(\Phi_i) \leq i \ \forall i\}$

$$\Phi_0 = 1, \qquad \Phi_{k+1} = x^{k+1} - \sum_{i=0}^{k} \frac{\langle x^{k+1}, \Phi_i \rangle}{\|\Phi_i\|^2} \Phi_i$$

Veamos que $\langle x\Phi_k, \Phi_j \rangle = 0$ si $j = 0, 1, \dots, k-2$.

$$\langle x\bar{\Phi}_k, \bar{\Phi}_j\rangle = \left\langle x\bar{\Phi}_k, x^j - \sum_{i=0}^{j-1}{}' \frac{\langle x^j, \bar{\Phi}_i\rangle}{\|\bar{\Phi}_i\|^2}\bar{\Phi}_i \right\rangle$$

$$= \langle x\bar{\Phi}_k, x^j\rangle - \sum_{i=0}^{j-1}{}' \frac{1}{\|\bar{\Phi}_i\|^2}\langle x\bar{\Phi}_k, \underbrace{\langle x^j, \bar{\Phi}_i\rangle}_{\#}\bar{\Phi}_i\rangle$$

$$= \langle x\bar{\Phi}_k, x^j\rangle - \sum_{i=0}^{j-1}{}' \frac{\langle x^j, \bar{\Phi}_i\rangle}{\|\bar{\Phi}_i\|^2}\langle x\bar{\Phi}_k, \bar{\Phi}_i\rangle$$

Fijando x.

$$\langle x\bar{\Phi}_k, \Phi_{k-2}\rangle = \left\langle x\left(x^k - \sum_{j=0}^{k-1}{}' \frac{\langle x^k, \bar{\Phi}_j\rangle}{\|\bar{\Phi}_j\|^2}\bar{\Phi}_j\right), \bar{\Phi}_{k-2}\right\rangle$$

$$= \left\langle x^{k+1} - \sum_{j=0}^{k-1}{}' \frac{\langle x^{k+1}, \bar{\Phi}_j\rangle}{\|\bar{\Phi}_j\|^2}\bar{\Phi}_j, \bar{\Phi}_{k-2}\right\rangle$$

$$= \left\langle \underbrace{x^{k+1} - \sum_{j=0}^{k}{}' \frac{\langle x^{k+1}, \bar{\Phi}_j\rangle}{\|\bar{\Phi}_j\|^2}\bar{\Phi}_j}_{\bar{\Phi}_{k+1}}, \bar{\Phi}_{k-2}\right\rangle + \frac{\langle x^{k+1}, \bar{\Phi}_k\rangle}{\|\bar{\Phi}_k\|^2}\langle \bar{\Phi}_k, \bar{\Phi}_{k-2}\rangle.$$

$$= \langle \bar{\Phi}_{k+1}, \bar{\Phi}_{k-2}\rangle + \frac{\langle x^{k+1}, \bar{\Phi}_k\rangle}{\|\bar{\Phi}_k\|^2}\langle \bar{\Phi}_k, \bar{\Phi}_{k-2}\rangle = 0.$$

En fonces:

$$x\bar{\Phi}_k(x) = \frac{\langle x\phi_k, \phi_{k+1}\rangle}{\|\phi_{k+1}\|^2}\phi_{k+1}(x) + \frac{\langle x\phi_k, \phi_k\rangle}{\|\phi_k\|^2}\phi_k(x) + \frac{\langle x\phi_k, \phi_{k-1}\rangle}{\|\phi_{k-1}\|^2}\phi_{k-1}(x)$$

Implica que

$$\bar{\Phi}_{k+1}(x) = \frac{\|\bar{\Phi}_{k+1}\|^2}{\langle x\bar{\Phi}_k, \bar{\Phi}_{k+1}\rangle}\left\{x - \frac{\langle x\bar{\Phi}_k, \bar{\Phi}_k\rangle}{\|\bar{\Phi}_k\|^2}\right\}\bar{\Phi}_k(x)$$

$$- \frac{\langle x\bar{\Phi}_k, \bar{\Phi}_{k-1}\rangle}{\langle x\bar{\Phi}_k, \bar{\Phi}_{k+1}\rangle}\frac{\|\bar{\Phi}_{k+1}\|^2}{\|\bar{\Phi}_{k-1}\|^2}\bar{\Phi}_{k-1}(x)$$

Sea $\alpha_k = \dfrac{\|\bar{\Phi}_{k+1}\|^2}{\langle x\bar{\Phi}_k, \bar{\Phi}_{k+1}\rangle}$, $\beta_k = \dfrac{\langle x\bar{\Phi}_k, \bar{\Phi}_k\rangle}{\|\bar{\Phi}_k\|^2}$, $\gamma_k = \dfrac{\langle x\bar{\Phi}_k, \bar{\Phi}_{k-1}\rangle}{\langle x\bar{\Phi}_k, \bar{\Phi}_{k+1}\rangle}\dfrac{\|\bar{\Phi}_{k+1}\|^2}{\|\bar{\Phi}_{k-1}\|^2}$.